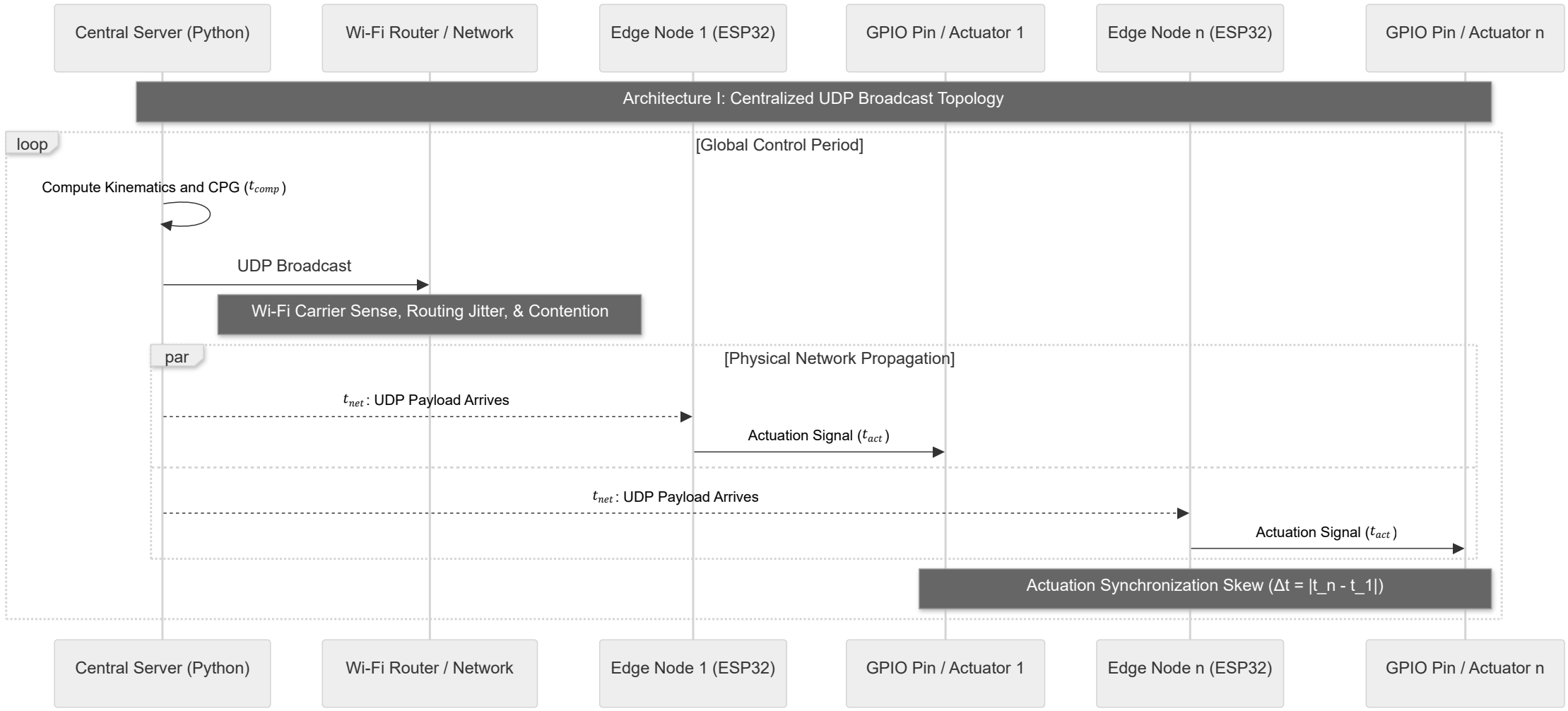
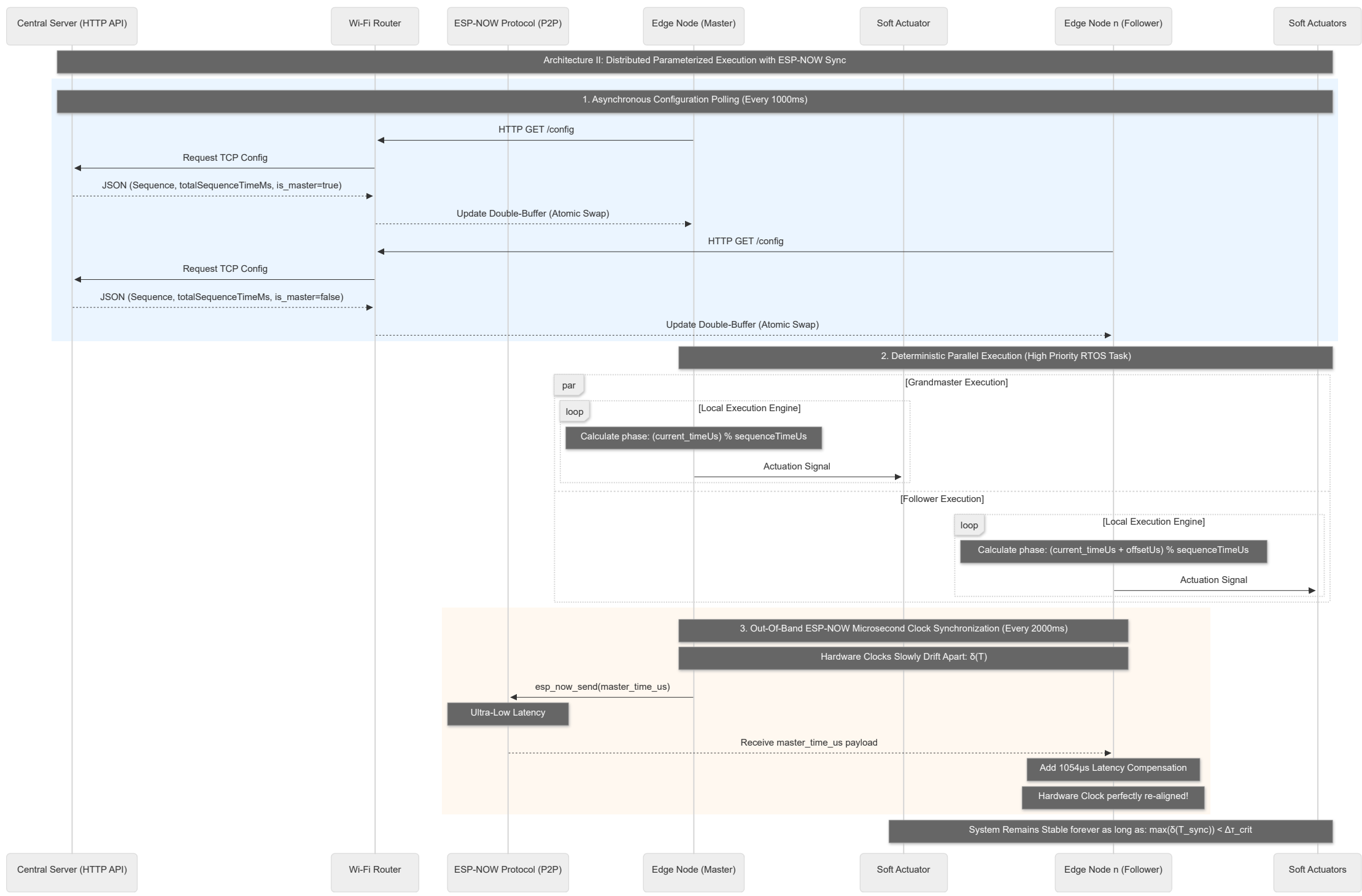
a)


Central Server (Python)
Wi-Fi Router / Network
Edge Node 1 (ESP32)
GPIO Pin / Actuator 1
Edge Node n (ESP32)
GPIO Pin / Actuator n
Architecture I: Centralized UDP Broadcast Topology
loop
[Global Control Period]
Compute Kinematics and CPG (t_{comp})
UDP Broadcast
Wi-Fi Carrier Sense, Routing Jitter, & Contention
par
[Physical Network Propagation]
t_{net}: UDP Payload Arrives
Actuation Signal (t_{act})
t_{net}: UDP Payload Arrives
Actuation Signal (t_{act})
Actuation Synchronization Skew (Δt = |t_n - t_1|)
Central Server (Python)
Wi-Fi Router / Network
Edge Node 1 (ESP32)
GPIO Pin / Actuator 1
Edge Node n (ESP32)
GPIO Pin / Actuator n

b)


Central Server (HTTP API)
Wi-Fi Router
ESP-NOW Protocol (P2P)
Edge Node (Master)
Soft Actuator
Edge Node n (Follower)
Soft Actuators
Architecture II: Distributed Parameterized Execution with ESP-NOW Sync
1. Asynchronous Configuration Polling (Every 1000ms)
HTTP GET /config
Request TCP Config
JSON (Sequence, totalSequenceTimeMs, is_master=true)
Update Double-Buffer (Atomic Swap)
HTTP GET /config
Request TCP Config
JSON (Sequence, totalSequenceTimeMs, is_master=false)
Update Double-Buffer (Atomic Swap)
2. Deterministic Parallel Execution (High Priority RTOS Task)
par
[Grandmaster Execution]
loop
[Local Execution Engine]
Calculate phase: (current_timeUs) % sequenceTimeUs
Actuation Signal
[Follower Execution]
loop
[Local Execution Engine]
Calculate phase: (current_timeUs + offsetUs) % sequenceTimeUs
Actuation Signal
3. Out-Of-Band ESP-NOW Microsecond Clock Synchronization (Every 2000ms)
Hardware Clocks Slowly Drift Apart: δ(T)
esp_now_send(master_time_us)
Ultra-Low Latency
Receive master_time_us payload
Add 1054µs Latency Compensation
Hardware Clock perfectly re-aligned!
System Remains Stable forever as long as: max(δ(T_sync)) < Δτ_crit
Central Server (HTTP API)
Wi-Fi Router
ESP-NOW Protocol (P2P)
Edge Node (Master)
Soft Actuator
Edge Node n (Follower)
Soft Actuators